Chi-Square Test

Entity 1: People's favorite colors

Entity 2: Preference for ice-cream

$H_0 \Rightarrow$ Favorite color and ice-cream preference are independent

$H_A \Rightarrow$ They are dependent

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

O → Observed Frequency

E → Expected Frequency

Categorical Variables

Classify data into distinct, non-numerical groups

Distinct Groups :- No overlap Eg :- Hair Color

Non-Numerical :- No inherit order Eg :- Apple ≠ Orange

Limited options :- Fixed categories Eg :- Traffic lights

Step 1. Define Hypothesis

H_0: Observed frequencies match the expected distribution.

H_A: Observed frequencies do not match the expected distribution.

Step 2. Table

	Chocolate	Vanilla	Strawberry	
Male	20	15	10	45
Female	25	20	30	75
Total	45	35	40	120

Step 3. Observed frequency:

$$E_{ij} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

Male and Chocolate $\Rightarrow \frac{45 \times 45}{120} = 16.875$

Male and Vanilla $\Rightarrow \frac{45 \times 35}{120}$

$\Rightarrow 13.125$

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	Chocolate	Vanilla	Strawberry
Male	16.875	13.125	15.0
Female	12.125	21.875	25.0

Step4. Perform Chi-Square Test on Male Category:

$$\chi^2 = \Sigma \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(20-16.875)^2}{16.875} + \frac{(15-13.125)^2}{13.125} + \frac{(10-15.0)^2}{15}$$

$$= \cancel{15.6898} + \cancel{11.98} + 1.667 \Rightarrow 14.947$$

$$9.765 + 3.515$$

Step5. DOF

df = (no. of rows −1) × (no. of colums −1)

$= (r-1) * (c-1)$

$= (2-1) * (3-1) \Rightarrow 2$

$X^2 = 14.947$ df=2

C.V at $\alpha = 0.05$ is 5.991

$14.947 < 5.991$

We reject null hypothesis

Goodness of Fit

Checks if hypothesized model matches observed data.

~~Step~~ Purpose: Validate if data fits an expected distribution.

Data Types: Work for both categorical and continuous data.

Benefits: Identifies model-data mismatch

Applications: Compare observed vs expected frequencies → Chi-Square

↓

Normal distribution

(3)

Sociologist conducted survey of 20 adults. She want to report the frequency distribution of the ages of the survey respondents.

Age in years: 52, 34, 32, 29, 63, 40, 46, 54, 36, 36, 24, 19, 45, 20, 28, 29, 38, 33, 49, 37

Grouped Frequency Distribution

range = Highest − Lowest = 63 − 19 = 44

$$\text{Width} = \frac{\text{range}}{\sqrt{\text{sample size}}} = \frac{44}{\sqrt{20}} = 9.84 = 10$$

Age		Frequency
$19 \le a < 29$	$\rightarrow$	4
$29 \le a < 39$	$\rightarrow$	9
$39 \le a < 49$	$\rightarrow$	3
$49 \le a < 59$	$\rightarrow$	3
$59 \le a < 69$	$\rightarrow$	1

Relative Frequency Distribution

Chikadee	3	$3/(3+1+4+2+4+2) = 3/16 = 0.19$
Dove	1	0.06
Finch	4	0.25
Grackle	2	0.13
Sparrow	4	0.25
Starling	2	0.13

Cumulative frequency

Age	Frequency	Cumulative Frequency	Cumulative Relative frequency
$19 \le a < 29$	4	4	0.2
$29 \le a < 39$	9	$9+4=13$	0.65
$39 \le a < 49$	3	$9+4+3=16$	0.8
$49 \le a < 59$	3	19	0.95
$59 \le a < 69$	1	20	1

Q. JFK airport hires you to estimate how the punctuality of flight arrivals is affected by weather. You begin by defining a probability space for which the sample space is:

$$\Omega = \{\text{late and rain, late and no rain, on time and rain, on time and no rain}\}$$

$P(\text{late, no rain}) = \frac{2}{20}$, $P(\text{on time, no rain}) = \frac{14}{20}$

$P(\text{late, rain}) = \frac{3}{20}$, $P(\text{on time, rain}) = \frac{1}{20}$

$$P(\text{late} \mid \text{rain}) = \frac{P(\text{late, rain})}{P(\text{rain})} = \frac{3/20}{3/20 + 1/20} = \frac{3}{4}$$

$P(\text{late} \mid \text{no rain}) = 1/8$

~~Suppose $S = \{1,2,3,4\}$ and $P(\{1,2\}) = 1/3$ \ $P(\{2,3\}) = 2/3$~~

~~Compute: $P(\{1\})$, $P(\{2\})$ and $P(\{3\})$~~

$P(\{1\}) = 1/12$, $P(\{1,2\}) = 1/6$ and $P(\{1,2,3\}) = 1/3$

Q: Suppose $S = \{1,2,3,4\}$,

compute $P\{1\}$, $P(\{2\})$, $P(\{3\})$, $P(\{4\})$

$\boxed{P1 = P(\{1\}) = 1/12}$

$P2 = P(\{1,2\}) - P(\{1\}) = \frac{1}{6} - \frac{1}{12} = \frac{2}{12}$

[Additivity property of probability:

$P(\{1,2\}) = P(\{1\}) + P(\{2\})$

1 & 2 are disjoint

$P(\{2\}) = P(\{1,2\}) - P(\{1\})$

Convert into denominator $\Rightarrow \frac{2}{12}$

$P2 = \frac{2}{12} - \frac{1}{12} = \frac{1}{12}$

$\boxed{P2 = \frac{1}{12}}$

$P_3 \Rightarrow \quad P(\{1,2,3\}) = \frac{1}{3}$

$P(\{1,2\}) = \frac{1}{6}$

Using Additivity property:

$$P(\{1,2,3\}) = P(\{1,2\}) + P(\{3\})$$

bcz union and they are disjoint

$$P_3 = P(\{1,2,3\}) - P(\{1,2\})$$

$$= \frac{1}{3} - \frac{1}{6} = \frac{2}{6} \text{ (Converting denominator 6)}$$

$$= \frac{2}{6} - \frac{1}{6} = \frac{1}{6}$$

$\boxed{P3 = \frac{1}{6}}$

$P_4 \Rightarrow \quad P_1 = \frac{1}{12}, \quad P_2 = \frac{1}{12}, \quad P_3 = \frac{1}{6}$

Space $S = \{1,2,3,4\}$

$$P_1 + P_2 + P_3 + P_4 = 1$$

$$P_4 = 1 - (P_1 + P_2 + P_3)$$

1. Add $P_1 + P_2$: $\frac{1}{12} + \frac{1}{12} = \frac{1}{6}$

2. Now Add P_3: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

3. Subtract from 1: $1 - \frac{1}{3} = \frac{2}{3}$

$P_1 = \frac{1}{12}$	$P_2 = \frac{1}{12}$	$P_3 = \frac{1}{6}$	$P_4 = \frac{2}{3}$